

Docker Container Monitoring using Cron Job

1. Create folders:

```
sudo mkdir -p /opt/container-monitor/logs
cd /opt/container-monitor
```

2. Create script:

```
sudo nano monitor.sh
```

3. Add monitoring script using docker stats and save.

4. Permission:

```
sudo chmod +x /opt/container-monitor/monitor.sh
```

5. Test:

```
sudo /opt/container-monitor/monitor.sh
```

6. Check log:

```
cat /opt/container-monitor/logs/container_usage.log
```

7. Cron:

```
sudo crontab -e
```

Add:

```
* * * * * /opt/container-monitor/monitor.sh
```

8. Verify:

```
sudo crontab -l
```

9. Check logs again:

```
cat /opt/container-monitor/logs/container_usage.log
```

10. Verify container:

```
docker ps
```

1.

```
ubuntu@ip-172-31-34-169:~$ ls
docker-project
ubuntu@ip-172-31-34-169:~$ sudo mkdir -p /opt/container-monitor/logs
ubuntu@ip-172-31-34-169:~$ cd /
ubuntu@ip-172-31-34-169:/$ ls
bin  boot  dev  etc  home  lib  lib64  lost+found  media  mnt  opt  proc  root  run  sbin  snap  srv  sys  tmp  usr  var
ubuntu@ip-172-31-34-169:/$ cd opt/
ubuntu@ip-172-31-34-169:/opt$ ls
container-monitor  containerd
ubuntu@ip-172-31-34-169:/opt$ cd container-monitor/
ubuntu@ip-172-31-34-169:/opt/container-monitor$ ls
logs
ubuntu@ip-172-31-34-169:/opt/container-monitor$ cd logs/
ubuntu@ip-172-31-34-169:/opt/container-monitor/logs$ sudo su
root@ip-172-31-34-169:/opt/container-monitor/logs# vi monitor.sh
root@ip-172-31-34-169:/opt/container-monitor/logs# cat monitor.sh
#!/bin/bash

LOG_FILE="/opt/container-monitor/logs/container_usage.log"

TIMESTAMP=$(date +%Y-%m-%d %H:%M:%S)

docker stats --no-stream --format "Container: {{.Name}}, CPU: {{.CPUPerc}}, Memory: {{.MemUsage}}" | while read line
do
    echo "$TIMESTAMP | $line" >> "$LOG_FILE"
done
root@ip-172-31-34-169:/opt/container-monitor/logs# sudo chmod +x /opt/container-monitor/monitor.sh
chmod: cannot access '/opt/container-monitor/monitor.sh': No such file or directory
root@ip-172-31-34-169:/opt/container-monitor/logs# sudo chmod +x monitor.sh
root@ip-172-31-34-169:/opt/container-monitor/logs# ls
monitor.sh
```

2.

```
bin boot dev etc home lib lib64 lost+found media mnt opt proc root run sbin snap srv sys tmp usr var
root@ip-172-31-34-169:/# cd opt/
root@ip-172-31-34-169:/opt# ls
container-monitor  containerd
root@ip-172-31-34-169:/opt# cd container-monitor/
root@ip-172-31-34-169:/opt/container-monitor# ls
logs
root@ip-172-31-34-169:/opt/container-monitor# cd logs/
root@ip-172-31-34-169:/opt/container-monitor/logs# ls
monitor.sh
root@ip-172-31-34-169:/opt/container-monitor/logs# chmod -x monitor.sh
root@ip-172-31-34-169:/opt/container-monitor/logs# mv monitor.sh /opt/container-monitor/logs
mv: 'monitor.sh' and '/opt/container-monitor/logs/monitor.sh' are the same file
root@ip-172-31-34-169:/opt/container-monitor/logs# mv monitor.sh /opt
root@ip-172-31-34-169:/opt/container-monitor/logs# cd ..
root@ip-172-31-34-169:/opt/container-monitor# ls
logs
root@ip-172-31-34-169:/opt/container-monitor# cd ..
root@ip-172-31-34-169:/opt# ls
container-monitor  containerd  monitor.sh
root@ip-172-31-34-169:/opt# mv monitor.sh /opt/container-monitor/
root@ip-172-31-34-169:/opt# cd container-monitor/
root@ip-172-31-34-169:/opt/container-monitor# ls
logs  monitor.sh
root@ip-172-31-34-169:/opt/container-monitor# chmod -x monitor.sh
root@ip-172-31-34-169:/opt/container-monitor# cd ..
root@ip-172-31-34-169:/opt# cd ..
root@ip-172-31-34-169:/# sudo /opt/container-monitor/monitor.sh
sudo: cannot execute '/opt/container-monitor/monitor.sh': Permission denied (os error 13)
root@ip-172-31-34-169:/# sudo chmod +x /opt/container-monitor/monitor.sh
sudo /opt/container-monitor/monitor.sh
root@ip-172-31-34-169:/# ls -l /opt/container-monitor/
total 8
drwxr-xr-x 2 root root 4096 Jul 11 10:10 logs
-rwxr-xr-x 1 root root 283 Jul 11 09:59 monitor.sh
root@ip-172-31-34-169:/# cat /opt/container-monitor/logs/container_usage.log
2026-07-11 10:10:54 | Container: myweb-container, CPU: 0.00%, Memory: 3.488MiB / 908.7MiB
2026-07-11 10:10:54 | Container: myweb-container, CPU: 0.00%, Memory: 3.488MiB / 908.7MiB
root@ip-172-31-34-169:/#
```

3.

```
crontab: installing new crontab
root@ip-172-31-34-169:/# sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
* * * * * /opt/container-monitor/monitor.sh
root@ip-172-31-34-169:/# cat /opt/container-monitor/logs/container_usage.log
2026-07-11 10:10:54 | Container: myweb-container, CPU: 0.00%, Memory: 3.488MiB / 908.7MiB
2026-07-11 10:15:01 | Container: myweb-container, CPU: 0.00%, Memory: 3.488MiB / 908.7MiB
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2026-07-11 10:16:01 | Container: myweb-container, CPU: 0.00%, Memory: 3.488MiB / 908.7MiB
2026-07-11 10:17:01 | Container: myweb-container, CPU: 0.00%, Memory: 3.984MiB / 908.7MiB
root@ip-172-31-34-169:/#
```

4.

```
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
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2026-07-11 10:17:01 | Container: myweb-container, CPU: 0.00%, Memory: 3.984MiB / 988.7MiB
root@ip-172-31-34-169:/# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
8592c7670707   myweb:v1   "/docker-entrypoint..." 6 hours ago    Up 6 hours    0.0.0.0:8000->80/tcp, [::]:8000->80/tcp  myweb-container
root@ip-172-31-34-169:/#
```